

Automated Classification of Sleep Paralysis Narratives: A Comparative Study of Western and South Asian Experiences Using Large Language Models

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Abstract

Sleep paralysis is a common parasomnia in which a person regains awareness while REM muscle atonia persists, often accompanied by vivid hallucinations. Prior research classifies these hallucinations using Cheyne et al.'s three factor taxonomy (Intruder, Incubus, and Vestibular Motor), but nearly all large scale narrative analysis of sleep paralysis has relied on Western, Reddit sourced data and manual qualitative coding. This study applies automated large language model classification to first person sleep paralysis narratives, comparing a Reddit sourced sample against a South Asian sample collected specifically for this research. A total of 151 filtered Reddit narratives and 16 filtered South Asian narratives, gathered through a targeted online survey, were classified using a zero shot prompting approach built on Cheyne's original category definitions. Intruder was the most common category in both samples, but the South Asian sample showed proportionally more Incubus and Vestibular Motor classifications. A distinct Auditory or Vibrational pattern emerged in both samples that Cheyne's original taxonomy does not capture. Recruitment for the South Asian sample proved considerably harder than for the Reddit sample, a pattern this paper argues reflects genuine discomfort discussing an experience bound up with jinn and supernatural meaning in South Asian cultural contexts, rather than a simple recruitment shortfall. This study contributes a reproducible method for automated, large scale sleep paralysis narrative classification, and a preliminary but suggestive account of how cultural framing may shape which hallucinatory themes are reported and how willingly they are shared.

1. Introduction

Sleep paralysis is a well documented parasomnia in which a person regains conscious awareness while the muscle atonia of REM sleep persists, leaving them temporarily unable to move or speak. Episodes are frequently accompanied by vivid hallucinations, commonly categorized using Cheyne et al.'s (1999) three factor taxonomy: the Intruder factor, involving a sensed presence or a perceived figure in the room, the Incubus factor, involving chest pressure and breathing difficulty, and the Vestibular Motor factor, involving floating, flying, or out of body sensations.

While sleep paralysis is a universal neurological phenomenon, how it is experienced, described, and explained varies substantially across cultures. Western accounts, particularly those shared online, tend to describe shadow figures or demonic entities. In many South Asian and broader Muslim contexts, the same physiological event is frequently attributed to jinn, and interpreted through religious and folk frameworks rather than purely medical ones. This cultural variation matters. It shapes how people cope with the experience, whether they seek medical help, and how distressing they find it.

Despite this, most large scale narrative analysis of sleep paralysis has relied on Western, predominantly Reddit sourced data, using manual, human coded qualitative methods rather than automated classification. Large language models now make it feasible to classify large volumes of first person narrative text against an established psychological taxonomy quickly and consistently. To our knowledge, this has not yet been applied to sleep paralysis narratives, nor used to directly compare Western and South Asian narrative samples.

This study addresses that gap and asks two questions. First, can an LLM reliably classify first person sleep paralysis narratives into Cheyne's Intruder, Incubus, and Vestibular Motor categories using a zero shot prompting approach? Second, how does the thematic distribution of these categories, and any patterns outside the established taxonomy, differ between a Western Reddit sourced sample and a South Asian, primarily Pakistani, sample collected specifically for this study?

2. Literature Review

The starting point for any sleep paralysis hallucination classification is Cheyne et al.'s (1999) factor analytic work, which established the Intruder, Incubus, and Vestibular Motor categories used throughout this paper. This taxonomy has remained the dominant framework in the field for over two decades and is the classification scheme adopted here.

Building on this framework, researchers have taken two broadly separate paths. One path has focused on prevalence and risk factors. Within Pakistan specifically, several cross-sectional surveys converge on similar conclusions. A study of 297 medical students across Karachi found a high proportion of respondents reporting sleep paralysis, with a significant association between episodes and academic impact (Muzammil et al., 2023). These prevalence studies establish that the population this paper draws its South Asian sample from experiences sleep paralysis at meaningful rates, but none of them analyze the content or structure of participants' narrative descriptions. They rely on structured survey instruments measuring frequency and correlates, not open ended narrative text.

A second path has focused on how sleep paralysis is culturally interpreted rather than how often it occurs. A substantial literature documents that it is explained through local supernatural frameworks worldwide, and within Islamic contexts specifically, it is frequently attributed by lay believers to jinn interference, with coping responses such as Quranic recitation and protective amulets commonly reported (Jalal et al., 2021). Most directly relevant to the present study is Monster of the Night (Khan et al., 2025), which examined gender based, religious, and cultural influences on how Pakistani university students narrate and interpret sleep paralysis through semi-structured interviews with 23 master's level students, finding that folklore and Islamic frameworks meaningfully shape the experience, with gender differences in which folk figures are invoked. This is the closest prior work to the present study in population and cultural scope, but it relies on qualitative interview based folk knowledge analysis rather than automated classification against an established taxonomy, and it does not compare its findings systematically to a Western narrative sample using a shared coding scheme.

A separate and more recent thread has begun applying computational methods to sleep paralysis. A doctoral thesis at the University of York conducted a qualitative thematic analysis of the r/SleepParalysis subreddit, manually coding narratives to understand how sufferers describe and make sense of their experiences online (Pryce Brooks, University of York). This demonstrates that Reddit is a viable and rich data source for sleep paralysis narrative research, though the coding itself remained manual. Separately, Ishaq et al. (2024) applied machine learning to structured survey data on sleep paralysis in a Pakistani sample, but their approach predicted demographic and behavioral correlates of occurrence rather than classifying the content of narrative text.

Where these paths do not yet meet is the specific combination this paper pursues: automated, LLM based classification of unstructured first person narrative text against Cheyne's established taxonomy, applied at a scale beyond what manual coding allows, and used to directly compare a Western Reddit sourced sample against a South Asian sample collected for this purpose. This is the gap the present study fills.

3. Methodology

3.1 Research Design

This study uses a mixed source, LLM assisted content classification design. Narratives were collected from two sources, a Reddit sample and a South Asian survey sample, filtered to isolate genuine first person accounts, and classified using an LLM against Cheyne's taxonomy, allowing systematic comparison between the two populations.

3.2 Reddit Sample Collection

First person sleep paralysis narratives were collected from the r/SleepParalysis subreddit using Reddit's public JSON endpoints, which return post data without requiring API authentication. This approach was adopted after two earlier methods failed. Reddit's official API, accessed through the PRAW library, could not be used because CAPTCHA verification during app registration repeatedly failed. A subsequent attempt to use unauthenticated requests directly from a cloud hosted notebook environment was blocked with HTTP 403 errors, most likely because Reddit rate limits traffic originating from data center IP ranges. The method that ultimately worked involved manually retrieving public JSON pages through a standard browser session and parsing them locally, which respected Reddit's public access terms while avoiding both obstacles.

Posts were collected across multiple paginated requests using Reddit's after cursor. An initial attempt using the top, all time sort order produced a pagination anomaly in which the returned posts began repeating earlier content after several pages, so the deduplicated dataset stopped growing despite the request log showing many more posts retrieved. Switching to the new sort order, which is strictly chronological rather than a ranked or cached listing, resolved this issue and allowed collection to continue growing across further rounds.

Each post's title and body text were concatenated into a single narrative unit. Posts under 30 words were excluded as too short to constitute a substantive narrative. This yielded 202 candidate narratives before further filtering.

3.3 South Asian Sample Collection

To address the absence of South Asian voices in existing Reddit sourced sleep paralysis literature, a supplementary sample was collected through an anonymous online survey, built using Jotform and distributed through the researcher's personal and university networks, including COMSATS University, the Microsoft Learn Student Ambassadors network, the Aspire Leaders Program alumni network, and social media, including a direct recruitment post on r/SleepParalysis itself.

The survey collected informed consent for anonymous academic use of responses, optional demographic information including age, gender, and self reported frequency, a required open ended prompt asking respondents to describe an episode in as much detail as they were comfortable sharing, and an optional question asking respondents to describe any cultural or religious beliefs or explanations they associate with their experience. This last question was included specifically to surface the South Asian cultural interpretation dimension motivating this study. The survey yielded 17 usable responses after removing duplicate submissions.

3.4 Data Filtering

Both datasets required filtering to isolate genuine first person experience narratives from adjacent but non narrative content. For the Reddit dataset, manual review excluded FAQ or informational posts, advice or tip posts, opinion or rant posts not describing a specific episode, recruitment posts for unrelated studies, subreddit moderation announcements, pure yes or no questions without narrative content, and posts describing unrelated phenomena such as general nightmare disorder or hypnic jerks. This filtering reduced the Reddit dataset from 202 to 151 narratives. For the South Asian dataset, one response describing auditory hallucination explicitly without an accompanying paralysis experience was removed, since it did not meet this study's definition of a sleep paralysis episode, leaving 16 narratives for analysis.

3.5 Classification Procedure

Filtered narratives were classified using an LLM, specifically the Groq API running the openai/gpt-oss-120b model, at a low temperature setting to encourage classification consistency. Classification used a structured zero shot prompting approach. The prompt supplied operational definitions of each Cheyne category derived directly from the original taxonomy, an additional Other category for narratives not clearly matching any established category, and an Intruder_Incubus category to accommodate the well documented co-occurrence of Intruder and Incubus elements within a single episode. The model returned a structured response containing both the assigned category and a short reasoning statement, allowing every classification decision to be audited after the fact. On parsing failure or API error, up to three retries were attempted per narrative. Given its small size, the South Asian sample was classified using the same category definitions and rubric applied directly by the researcher rather than through the API pipeline, for practical reasons; this difference in classification procedure between the two samples is noted here for transparency and returned to in the limitations.

3.6 Post-hoc Sub-theme Analysis

Narratives classified as Other were manually reviewed to determine whether they reflected genuine patterns outside Cheyne's taxonomy rather than simple classification failure. This review identified an Auditory or Vibrational cluster, narratives centered on buzzing, ringing, or vibration without a clear visual intruder or chest pressure element, and a smaller Somatic or Physical cluster, narratives centered on pain, electric shock sensations, or partial paralysis. Narratives that fit neither sub-theme and lacked sufficient descriptive detail for confident classification were retained and labeled Vague or Unclear, in keeping with the decision to preserve the full filtered dataset rather than exclude low detail narratives after the fact.

3.7 Researcher Positionality

The researcher has personal lived experience of recurrent sleep paralysis, which was the original motivation for this project. This experience was not treated as a data point in either analyzed sample. It is disclosed here in the interest of methodological transparency, and in keeping with common practice in qualitative research on culturally embedded experiences, where a researcher's own relationship to the phenomenon can shape both the questions asked and the sensitivity brought to participants' accounts.

3.8 Ethical Considerations

Reddit data consisted exclusively of publicly accessible posts. No private or authentication gated content was accessed, and no usernames or other directly identifying information were retained in the analyzed dataset. Survey participation was voluntary and anonymous, with explicit consent obtained before any narrative was recorded, and no personally identifying information was collected at any point.

4. Results

4.1 Reddit Sample Classification

Of the 151 filtered Reddit narratives, the LLM classified 74 (49.0%) as Intruder, 8 (5.3%) as Incubus, 7 (4.6%) as Vestibular Motor, 4 (2.6%) as Intruder_Incubus, and 58 (38.4%) as Other.

Within the Other category, sub-theme review identified 12 narratives, 20.7% of Other and 7.9% of the full sample, reflecting the Auditory or Vibrational pattern; 5 narratives, 8.6% of Other and 3.3% of the full sample, reflecting the Somatic or Physical pattern; and the remaining 41 narratives, 70.7% of Other, classified as Vague or Unclear due to insufficient narrative detail.

4.2 South Asian Sample Classification

A supplementary South Asian sample was collected through the anonymous survey described in section 3.3, distributed through the researcher's personal network and a direct recruitment post on r/SleepParalysis. This effort yielded a much smaller sample than the Reddit collection, discussed further in section 5. After removing one duplicate submission and one response describing auditory hallucination without an accompanying paralysis experience, 16 usable narratives remained.

Of these 16 filtered South Asian narratives, classification against the same taxonomy identified 6 (37.5%) as Intruder, 2 (12.5%) as Incubus, 2 (12.5%) as Vestibular Motor, 1 (6.2%) as Intruder_Incubus, and 5 (31.2%) as Other. Within the Other category, 2 narratives reflected the

Auditory or Vibrational sub-theme also observed in the Reddit sample, and the remaining 3 were classified as Vague or Unclear.

Table 1 places both distributions side by side.

Category	Reddit sample (n=151)	South Asian sample (n=16)
Intruder	49.0% (74)	37.5% (6)
Other	38.4% (58)	31.2% (5)
Incubus	5.3% (8)	12.5% (2)
Vestibular-Motor	4.6% (7)	12.5% (2)
Intruder_Incubus	2.6% (4)	6.2% (1)

4.3 Cultural Interpretation Findings

Of the South Asian respondents who answered the optional cultural belief question, the majority indicated yes, that a cultural or religious explanation was associated with their experience, but most did not elaborate beyond this single word despite the question inviting further detail. One respondent provided a detailed account, attributing her experience to a *saya*, meaning shadow or presence, or a *jinn*, and described common protective responses including *Ruqyah Shariah*, the recitation of Quranic verses for spiritual protection, and the use of *taweez*, protective amulets. The brevity of most other responses to this question is itself notable and is discussed further in section 5.

4.4 Cross-Sample Comparison

The two samples show a broadly similar overall pattern, with Intruder as the most common category in both (49.0% Reddit, 37.5% South Asian) and Other as the second most common (38.4% Reddit, 31.2% South Asian). The most notable difference sits in the Incubus and Vestibular Motor categories, each of which appeared roughly two and a half times more often, proportionally, in the South Asian sample (12.5% each) than in the Reddit sample (5.3% and 4.6% respectively). This suggests physical and bodily sensations, chest pressure or choking in the Incubus case, floating or bodily displacement in the Vestibular Motor case, may be relatively more salient in how South Asian respondents in this sample described their experiences. Given the very small South Asian sample size, this pattern should be treated as suggestive rather than confirmed, a point returned to in section 5.

5. Discussion

The finding that Intruder was the most common category in the Reddit sample is broadly consistent with prior descriptions of sleep paralysis hallucinations as centering on a sensed threatening presence, and it aligns with the general pattern reported in the original Cheyne factor analysis.

The emergence of a distinct Auditory or Vibrational sub-theme within narratives that did not fit Cheyne's original taxonomy is a notable finding on its own. Buzzing, ringing, and vibration sensations were clearly reported by a meaningful number of sleep paralysis experiencers in both samples, and may be under-represented in a taxonomy developed primarily from earlier data collection methods. This does not necessarily call for a wholly new hallucination category. It does suggest that auditory and vibrational phenomena deserve more explicit attention in future narrative coding schemes, including this one.

The cross-sample comparison offers a preliminary but suggestive answer to this study's second research question. Intruder remained dominant in both samples, consistent with the broader literature's emphasis on sensed presence as the most commonly reported hallucinatory theme. The South Asian sample, however, showed proportionally more Incubus and Vestibular Motor classifications. Read alongside the cultural interpretation finding in section 4.3, in which the one detailed response explicitly described a felt physical presence, a jinn sitting or pressing, in a manner consistent with the Incubus category, this raises the possibility that South Asian religious framing may attach itself more readily to bodily, physically felt sensations than to purely visual intruder experiences. This would sit comfortably alongside *Monster of the Night*, which similarly found that Pakistani students' religious and folkloric framings shaped how physical sensations during sleep paralysis were interpreted (Khan et al., 2025). Given the sample size here, this connection is best treated as a hypothesis for future work rather than a confirmed finding.

A second and unexpected finding emerged from the data collection process itself, not from the narrative content. Recruitment for the South Asian sample was markedly more difficult than anticipated, yielding only 16 usable narratives despite active outreach across multiple personal and institutional networks, compared to 151 narratives obtained from Reddit's public posts alone. Several potential participants who were approached directly indicated discomfort with the topic, particularly given its association with jinn and supernatural explanation in the cultural contexts this study aimed to sample. If this pattern reflects a broader tendency rather than being specific to this study's recruitment channels, it suggests that the same cultural framing that makes South Asian sleep paralysis narratives analytically interesting, their embeddedness in religious and supernatural meaning, may also be precisely what makes people less willing to discuss the experience openly, even anonymously. This is worth reporting as a finding in its own right, not merely worked around

as a limitation, since it points to a structural difficulty in studying culturally sensitive experiences that future researchers in this area should plan for.

A methodological point also merits discussion. The data collection obstacles encountered here, CAPTCHA failures with Reddit's official API and rate limiting of cloud hosted requests, are not just logistical footnotes. They reflect a broader accessibility barrier that likely shapes which researchers are able to conduct large scale social media based narrative research, and are worth noting as a practical contribution alongside the substantive findings.

5.1 Limitations

The South Asian sample ($n=16$) is far smaller than the Reddit sample ($n=151$), and this asymmetry is the study's central limitation. As discussed above, this does not appear to be primarily a failure of recruitment effort but rather a reflection of genuine reluctance to discuss a topic bound up with supernatural and religious meaning, even under conditions of full anonymity. Because of this, the cross-sample comparison in section 4.4 should be read as an exploratory, hypothesis generating pattern rather than a statistically powered test of difference between populations, and no formal significance testing was conducted given the small cell sizes involved. Most South Asian respondents who indicated a cultural or religious association with their experience did not elaborate beyond a single word answer, which limited the depth of qualitative material available for section 4.3 and may itself be a further sign of the discomfort discussed above.

The two samples were also classified through slightly different procedures. The Reddit sample was classified through the Groq API pipeline described in section 3.5, while the smaller South Asian sample was classified directly by the researcher using the same category definitions. This was a practical choice given the South Asian sample's small size, but it introduces a consistency limitation that should be addressed in future work by running both samples through an identical automated pipeline.

LLM based classification, while enabling scale, has not been validated in this study against independent human inter-rater coding. Future work should address this by having a subset of narratives independently coded by human raters and comparing agreement rates. Finally, both samples are self-selecting. Reddit users who post about sleep paralysis and survey respondents willing to complete an anonymous form are not necessarily representative of all experiencers in their respective populations, and the South Asian sample in particular, given its small size and recruitment through the researcher's own networks, cannot be assumed to represent Pakistani or South Asian experiencers broadly.

6. Conclusion and Future Work

This study applied LLM based automated classification to first person sleep paralysis narratives, comparing a Reddit sourced Western sample against a South Asian sample collected specifically to address a gap in existing literature. Three findings stand out. Intruder was the dominant hallucination category in both samples, consistent with prior literature. A recurring Auditory or Vibrational pattern emerged outside Cheyne's original taxonomy in both samples, suggesting this dimension deserves more explicit attention in future coding schemes. The South Asian sample, though small, showed proportionally more Incubus and Vestibular Motor classifications than the Reddit sample, a pattern worth pursuing with a larger dataset, and the difficulty of recruiting that sample in the first place turned out to be a finding in itself, pointing to real reluctance to discuss a stigmatized, religiously meaningful experience even anonymously.

Future work could expand the South Asian sample through in-person or trusted-network recruitment to overcome the response rate limitations encountered here, validate LLM classification against human inter-rater coding, run both samples through an identical automated classification pipeline, and extend this comparative approach to additional cultural contexts beyond Pakistan to build a broader cross-cultural picture of how sleep paralysis is narratively constructed around the world.

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